MECH321 Dynamics of Engineering Systems

Lecture 7
Feedback

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TYUT · 2026

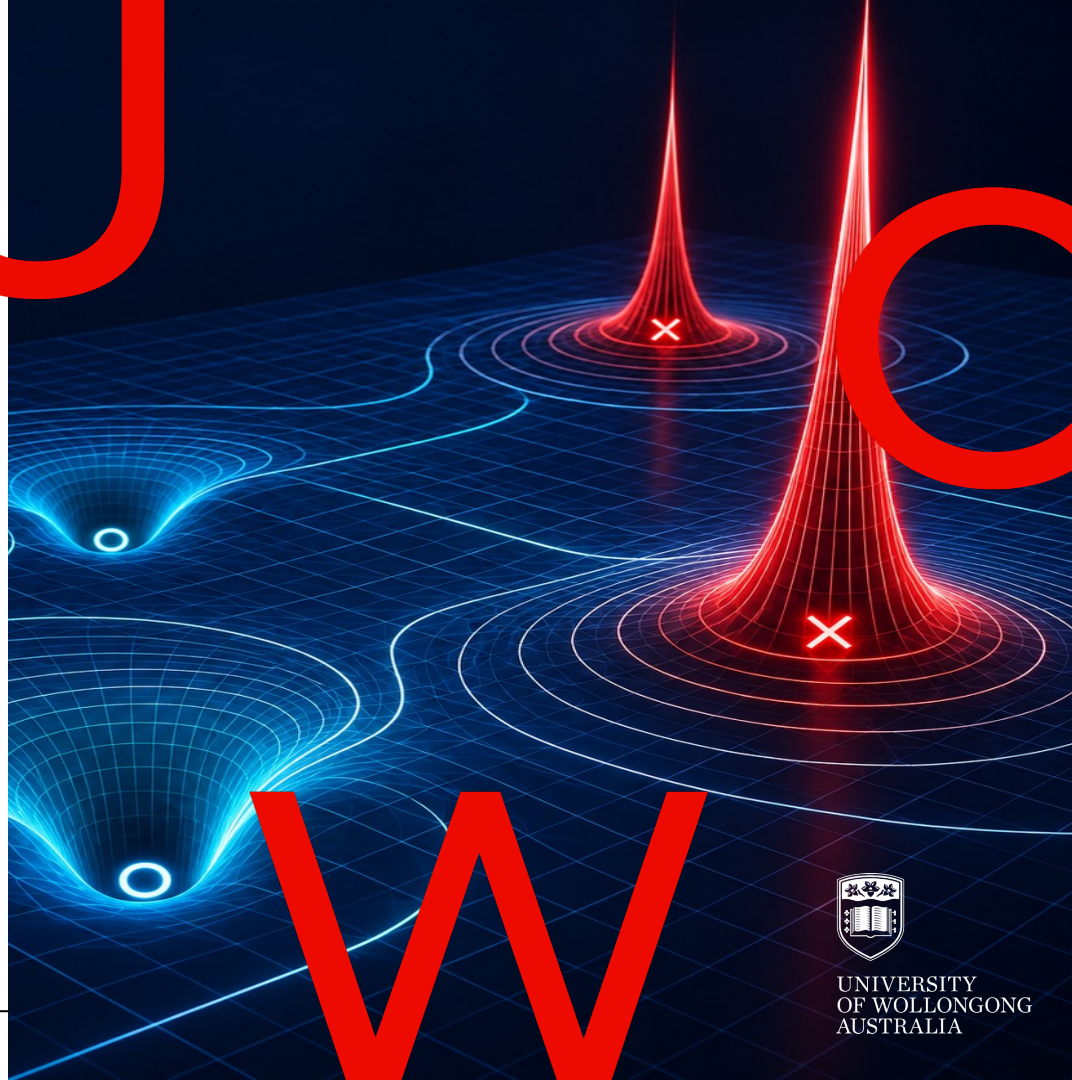


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PART 1

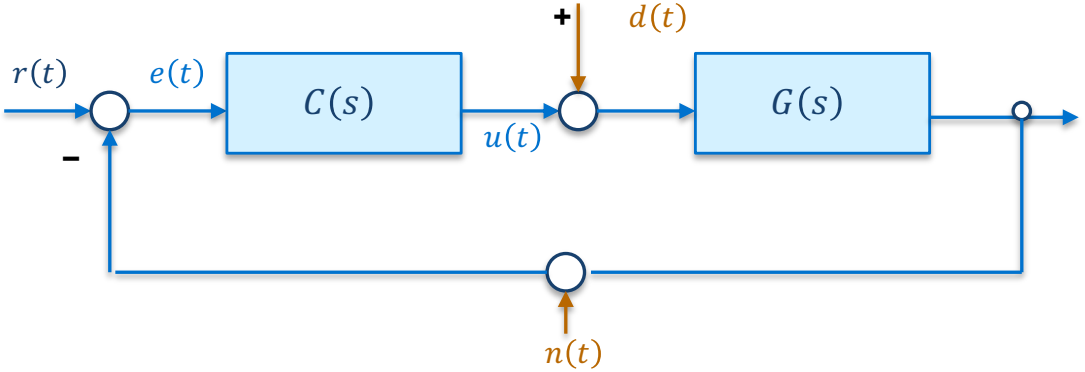
Feedback models and transient response

01 Loop signals

02 Reference and load paths

03 Poles, damping and overshoot

The feedback loop



$r(t)$ **reference**
$e(t)$ measured error
$u(t)$ control action
$d(t)$ **input load disturbance**
$n(t)$ **sensor noise**
$y(t)$ actual motor angle

Three external time signals: $r(t)$, $d(t)$, $n(t)$
Plant input $u(t) + d(t)$; measured output $y(t) + n(t)$; unity-feedback loop $L(s) = C(s)G(s)$.

Open-loop and closed-loop transfer functions

CORE

Unity negative feedback; each transfer path uses zero initial conditions

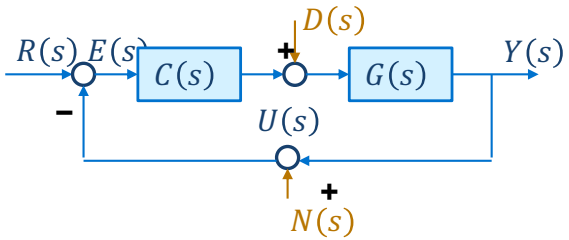
$L(s) = C(s)G(s)$ **Loop / open-loop TF (unity feedback)**

With sensor H: $L(s) = C(s)G(s)H(s)$

$T(s) = \dfrac{L(s)}{1 + L(s)}$ **Closed-loop reference TF**

Complementary sensitivity

$S(s) = \dfrac{1}{1 + L(s)}$ **Sensitivity / reference-to-error TF**



Path	Transfer function	Desired behaviour
reference $R(s)$ → output $Y(s)$	$T(s) = Y(s)/R(s)$	near 1 at low frequency
reference $R(s)$ → error $E(s)$	$S(s) = E(s)/R(s)$	small at low frequency
input load $D(s)$ → output $Y(s)$	$G(s)S(s) = Y(s)/D(s)$	small in the disturbance band
sensor noise $N(s)$ → output $Y(s)$	$-T(s) = Y(s)/N(s)$	small at high frequency

$S(s)+T(s) = 1$: tracking, disturbance rejection and noise sensitivity must be judged by frequency.

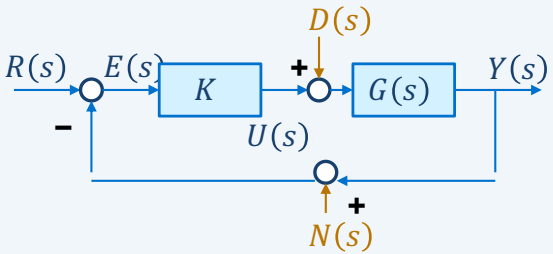
Position loop A: reference to angle

EXAMPLE

A motor obeys $\ddot{q}(t) + 2\dot{q}(t) = u(t) + d(t)$, with $y(t) = q(t)$ in rad and $u(t)$, $d(t)$ in rad/s².

Controller: $u(t) = K[r(t) - y_m(t)]$; measurement: $y_m(t) = y(t) + n(t)$.

Assume zero initial conditions. Set $d(t) = n(t) = 0$. Derive $Y(s)/R(s)$, then use $K = 4$ s⁻².



Solution

For the reference path, $D(s) = N(s) = 0$:

$$(s^2 + 2s)Y(s) = U(s), \quad G(s) = \frac{1}{s(s+2)}$$

$$Y(s) = G(s)U(s), \quad U(s) = K[R(s) - Y(s)]$$

$$[s(s+2) + K]Y(s) = KR(s)$$

$$\frac{Y(s)}{R(s)} = \frac{K}{s^2 + 2s + K}$$

$$K = 4\,\text{s}^{-2}: \quad T(s) = \frac{4}{s^2+2s+4}$$

Position loop A: is the closed loop stable?

QUESTION

For the same ideal motor, $K = 4$ s⁻² and $D(s) = N(s) = 0$. The reference transfer function has already been derived:

$$T(s) = \frac{4}{s^2 + 2s + 4}, \qquad s^2 + 2s + 4 = 0$$

1 Find both closed-loop poles. Complete the square or use the quadratic formula; include units.

2 Mark the poles in the s plane. Is the response decaying, growing or sustained? Will it oscillate?

3 Explain why a **negative summing sign alone is not a stability test** for an arbitrary plant.

Use the pole criterion (Lecture 3) and second-order form (Lecture 4).

Position loop A: poles prove stability

ANSWER

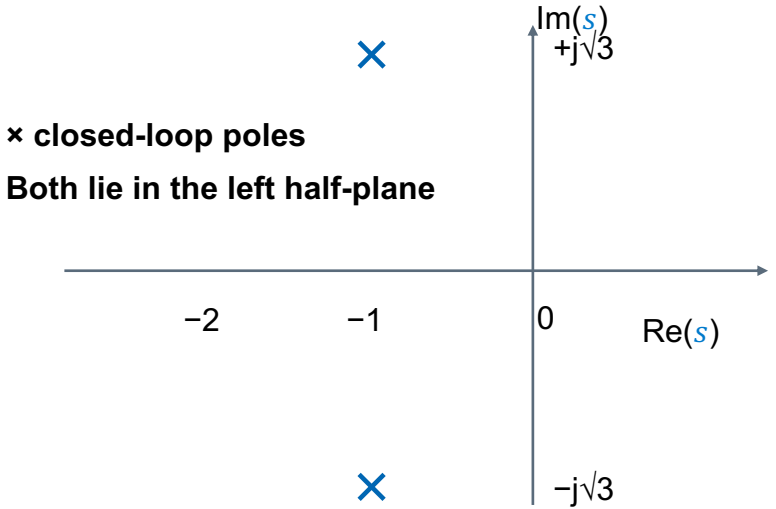
Complete the square

$$s^2 + 2s + 4 = (s+1)^2 + 3 = 0$$

$$s + 1 = \pm j\sqrt{3}$$

$$p_{1,2} = -1 \pm j\sqrt{3} \text{ s}^{-1}$$

Real part -1 s⁻¹: decay e^{-t}; imaginary parts $\pm\sqrt{3}$ s⁻¹: oscillation. Stable and underdamped.



GENERAL RULE For an uncancelled rational LTI model, all closed-loop poles must have Re(p)<0 for asymptotic stability. A right-half-plane pole gives a growing mode; imaginary-axis poles require separate care.

Result for $K = 4$ only. Negative feedback alone does not guarantee stability.

Position loop A: predicted step response

EXAMPLE

Match coefficients

$$T(s) = \frac{4}{s^2 + 2s + 4} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \qquad \omega_n^2 = 4, \quad 2\zeta\omega_n = 2 \quad \Rightarrow \quad \omega_n = 2 \text{ rad/s}, \quad \zeta = 0.5$$

$$\omega_d = \omega_n\sqrt{1 - \zeta^2} = \sqrt{3} \text{ rad/s}$$

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} = 0.163, \quad t_p = \frac{\pi}{\omega_d} = 1.81 \text{ s}$$

$$t_s\ (2\%) \approx \frac{4}{\zeta\omega_n} = 4.0 \text{ s}, \qquad T(0) = 1$$

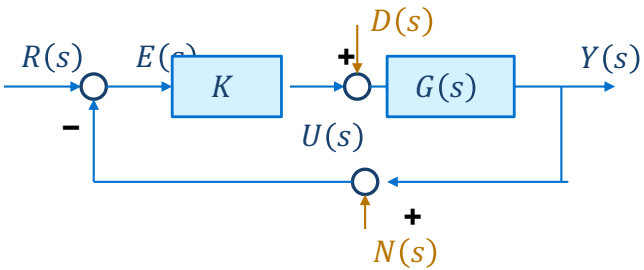
A 1 rad reference step settles to 1 rad, peaks at about 1.163 rad and has a decaying oscillation. The settling-time formula is an approximation, not an exact last crossing.

For $K = 9$, poles $-1 \pm j\sqrt{8}$ s⁻¹ give faster oscillation, but the decay exponent remains -1 s⁻¹. **More gain does not mean faster decay**.

Position loop B: load disturbance to angle

EXAMPLE

Ideal motor; output is angle. Compare the same load with the controller off and on.



Step load: 1 rad/s². Reference, sensor noise and initial conditions: zero.

$$K = 4$$

$$G(s) = \frac{1}{s(s+2)} \qquad D(s) = \frac{1}{s}, R(s) = 0, N(s) = 0$$

$$\frac{Y(s)}{D(s)} = \frac{G(s)}{1 + K\,G(s)} = \frac{1}{s^2 + 2s + K}$$

No control $\quad Y_{0(s)} = \dfrac{1}{s^2(s+2)} \qquad y_{0(t)} = \dfrac{t}{2} - \dfrac{1}{4} + \dfrac{e^{-2t}}{4} \to \infty$

With feedback $\quad Y(s) = \dfrac{1}{s(s^2 + 2s + 4)} \qquad y_\infty = \lim_{s \to 0} s\,Y(s) = \dfrac{1}{4}$

PART 2

Open-loop response and closed-loop stability

Stability from loop response $L(s)$

01 Characteristic equation

02 Bode and Nyquist boundary

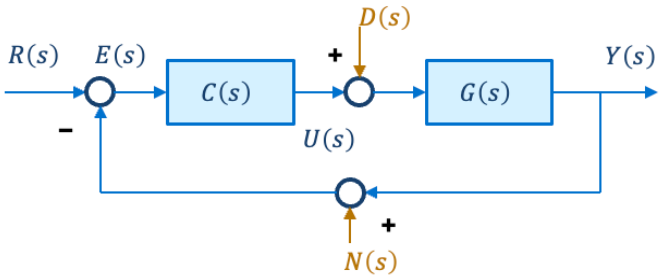
03 Verify with closed-loop poles

Open-loop data, closed-loop stability

Open-loop and closed-loop transfer functions

$$L(s) = C(s)G(s)$$

$$T(s) = \frac{Y(s)}{R(s)} = \frac{L(s)}{1 + L(s)}$$



For an uncancelled model, this denominator gives the closed-loop poles.

$$\Delta(j\omega) = 0 \iff 1 + L(j\omega) = 0 \iff L(j\omega) = -1$$

An imaginary-axis closed-loop pole requires the loop response to reach (−1, 0).

Next: locate this condition on Bode and Nyquist plots.

Critical Feedback Condition: Bode

$L(j\omega_*) = -1$

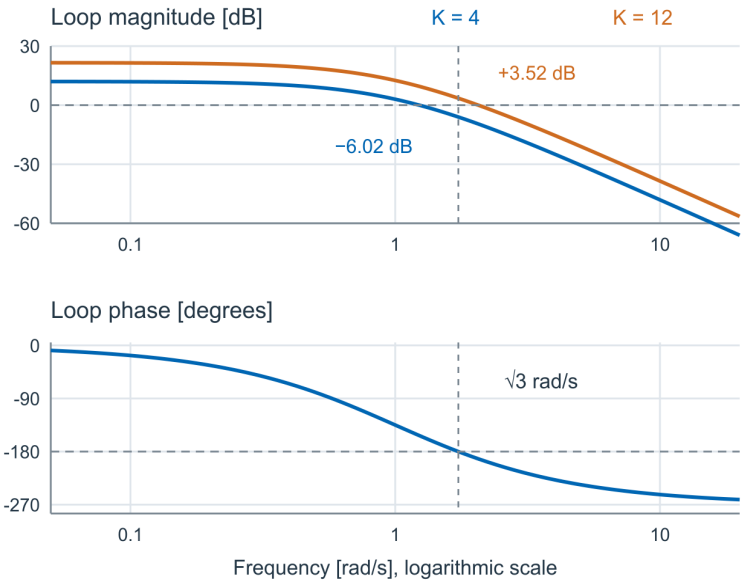
Two conditions must occur together:

Magnitude 1 = 0 dB
Phase −180° (modulo 360°)

Separate example: three stable lag modes

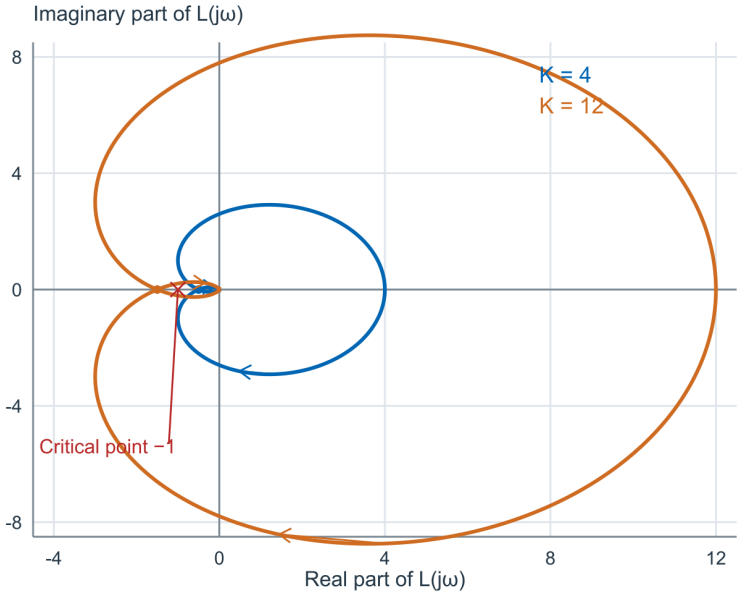
$$L_b(s) = \frac{K}{(s+1)^3}$$

$$\left|L_b\left(j\sqrt{3}\right)\right| = \frac{K}{(1+3)^{3/2}} = \frac{K}{8}$$



At √3 rad/s the phase is −180°. $K = 4$ gives −6.02 dB; $K = 12$ gives +3.52 dB.
Boundary: $K = 8$.

Critical Feedback Condition: Nyquist



Why the point −1?

$$F(s) = 1 + L(s), \quad L = -1 \iff F = 0$$

At the stability boundary:
$1 + L(j\omega) = 0$; critical point L = −1.

Here P=0:

$K = 4$: no encirclement → stable closed loop.
$K = 12$: encirclement → unstable closed loop.

Lecture 8 adds RHP-pole counting and imaginary-axis detours.

Worked check: the same boundary from the poles

For $L_b(s) = \frac{K}{(s+1)^3}$, find the gain at which the closed-loop poles reach the imaginary axis.

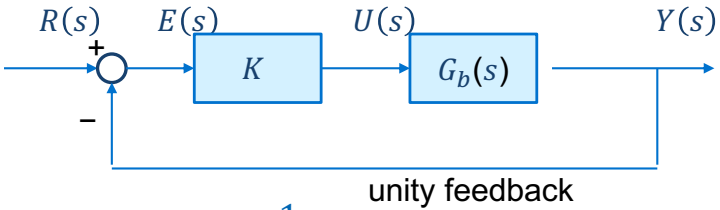
For this different plant, solve the closed-loop boundary by setting $s = j\omega$ ($\omega > 0$).

$$(s+1)^3 + K = s^3 + 3s^2 + 3s + 1 + K = 0$$

$$(1 + K - 3\omega^2) + j(3\omega - \omega^3) = 0$$

$$3\omega - \omega^3 = 0 \;\Rightarrow\; \omega = \sqrt{3} \quad (\omega > 0)$$

$$K = 8: \quad s^3 + 3s^2 + 3s + 9 = (s+3)(s^2+3)$$



$$G_b(s) = \frac{1}{(s+1)^3}$$

$$1 + K - 3(3) = 0 \;\Rightarrow\; K_{\text{crit}} = 8$$

$K = 4: -2.587; -0.206 \pm j1.375$

All poles are in the **left half-plane: stable**.

$K = 12: -3.289; +0.145 \pm j1.983$

Two poles are in the **right half-plane: unstable**.

PART 3

Tracking, disturbances and design tradeoffs

01 Tracking error and system type

02 Load offsets and noise

03 Gain tradeoffs

Steady-state error

CORE

The difference between the reference and actual output as time approaches infinity

1 Begin with reference minus output

$$Y(s) = T(s)R(s)$$

$$E(s) = R(s) - Y(s) = = [1 - T(s)]R(s)$$

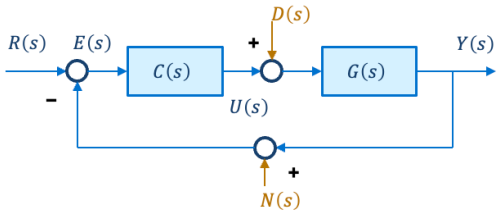
$$E(s) = \left(1 - \frac{L(s)}{1+L(s)}\right) R(s) = \frac{R(s)}{1+L(s)}$$

$$R(s) = \frac{A}{s} \quad \text{(step of height } A\text{)}$$

$$e_\infty = \lim_{t\to\infty} e(t) = \lim_{s\to 0} s\,E(s) = \lim_{s\to 0} \frac{A}{1+L(s)}$$

2 Shortcut for unity negative feedback

$$\frac{E(s)}{R(s)} = \frac{1}{1+L(s)}$$



Final-value theorem: every pole of sE(s) must lie in the left half-plane. **Check this before taking the limit**.

Always multiply $E(s)$ by s before taking $s \to 0$. For a step, this s cancels the input's *1/s*.

A noncancelled 1/s loop integrator gives zero step error: the closed loop must be stable and the remaining DC gain finite and nonzero.

System type: step-error derivations

EXAMPLE

Unity negative feedback, zero initial conditions, and no disturbance or sensor noise

DEFINITION Type N is the **number of uncancelled poles** at $s = 0$ in the loop transfer function $L(s) = C(s)G(s)$. Each such pole is a pure $1/s$ integrator.

QUESTION For $r(t) = 1$, $R(s) = 1/s$. Find each loop's steady-state step error.

Type 0: no integrator

$$L_0(s) = \frac{8}{s+2}$$

$$1 + L_0(s) = \frac{s+10}{s+2}$$

$$E_0(s) = \frac{s+2}{s(s+10)}$$

$$e_\infty = \lim_{s\to 0} s\,E_0(s) = \frac{2}{10} = 0.20$$

Type 1: one integrator

$$L_1(s) = \frac{8}{s(s+2)}$$

$$1 + L_1(s) = \frac{s^2+2s+8}{s(s+2)}$$

$$E_1(s) = \frac{s+2}{s^2+2s+8}$$

$$e_\infty = \lim_{s\to 0} \frac{s(s+2)}{s^2+2s+8} = 0$$

Stable closed-loop poles: -10 and $-1 \pm j\sqrt{7}$ s⁻¹ respectively. Both final-value calculations are valid.

System type: tracking-error summary

OPTIONAL

Optional unit-input reference: step $r(t) = 1$, ramp $r(t) = t$, parabola $r(t) = t^2/2$

Type	Integrators in $L(s)$	Step error	Ramp error	Parabola error
0	none	$\frac{1}{1 + K_p}$	∞	∞
1	one	0	$\frac{1}{K_v}$	∞
2	two	0	0	$\frac{1}{K_a}$

$$K_p = \lim_{s \to 0} L(s) \qquad K_v = \lim_{s \to 0} s\,L(s) \qquad K_a = \lim_{s \to 0} s^2\,L(s)$$

The position error constant is a limit of the whole loop, not the proportional gain K used in this feedback example.

Stable unity-feedback loops only. Count integrators in $L(s)$, **not plant order**. Step tracking and load rejection are different.

Position loop steady-state example A: tracking

Angle $y(t)$ [rad]; gain $K = 4$ s⁻².

Task: find $e_\infty = \lim_{t\to\infty}[r(t) - y(t)]$ for two separate commands.

(a) Move to and hold 1 rad: $r(t) = 1$ rad. **(*b*) Move continuously at 0.4** rad/s: $r(t) = 0.4\ t$ rad.

Both start at $t = 0$ from rest; $d(t) = n(t) = 0$. In (b), the output keeps moving: seek the final position lag.

$$\frac{E(s)}{R(s)} = S(s) = \frac{s(s+2)}{s^2 + 2s + 4}$$

Solution (a): fixed position

$$R(s) = \frac{1}{s}, \quad E(s) = \frac{s+2}{s^2 + 2s + 4}$$

$$e_\infty = \lim_{s\to 0} \frac{s(s+2)}{s^2 + 2s + 4} = 0$$

Solution (*b*): constant speed

$$R(s) = \frac{0.4}{s^2}, \quad E(s) = \frac{0.4(s+2)}{s(s^2 + 2s + 4)}$$

$$e_\infty = \lim_{s\to 0} \frac{0.4(s+2)}{s^2 + 2s + 4}$$

$$K_v = \lim_{s\to 0} s\,L(s) = \frac{K}{2} = 2$$

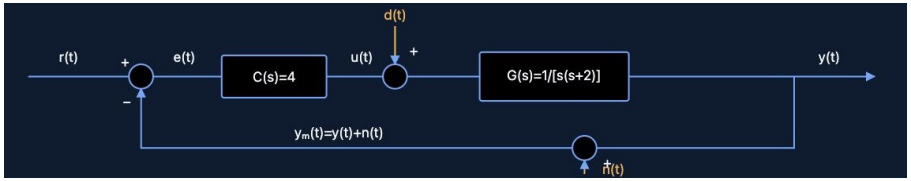
$$e_{\infty,\text{ramp}} = \frac{0.4}{K_v} = 0.20$$

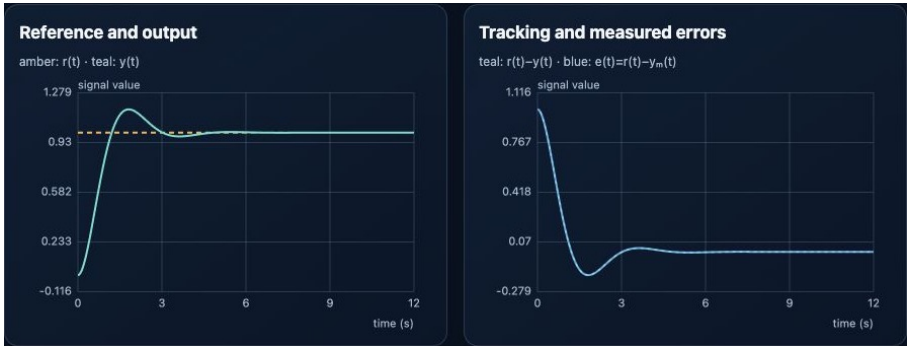
The stable poles are $-1 \pm j\sqrt{3}$ s⁻¹. The 0.20 rad result is a **position lag, not a speed error**.

Live demo: reference tracking

Reference tests at $K = 4$

OPEN LIVE DEMO





Feedback Explorer: Reference tracking

START Demo 6: Reference tracking. $K_c = 4$, $a = 2$, no load/noise.

1 Read the 1 rad step response and error. Then open the loop. Does $y(t)$ still settle?

2 Close it. Choose Ramp, slope 0.4 rad/s. Predict the long-term position lag before plotting.

MATLAB alternative (Control System Toolbox)

```
s=tf('s'); K=4; G=1/(s*(s+2));
T=feedback(K*G,1); t=(0:.01:12)';
figure; step(T,t); grid on
r=.4*t; y=lsim(T,r,t);
figure; plot(t,[r,y,r-y]); grid on
legend('r(t)','y(t)','r(t)-y(t)')
```

CHECK Closed loop: poles $-1 \pm j\sqrt{3}$ s^{-1}, overshoot 16.3%, step error 0. Ramp slope 0.4 rad/s gives final position lag 0.20 rad.

OPTIONAL

Position loop steady-state example B: constant load

Task: keep $r(t) = 0$ rad. At $t = 0$, apply $d(t) = 1$ rad/s² between the controller and plant. Use $K = 4$ s⁻², zero initial conditions and $n(t) = 0$. Find $Y(s)/D(s)$, then y_∞, e_∞ and u_∞.

1 Load path (slide 10)

$$E(s) = -Y(s), \quad U(s) = -4Y(s)$$

$$\frac{Y(s)}{D(s)} = \frac{G(s)}{1 + 4G(s)}$$

2 Steady values

$$Y(s) = \frac{1}{s(s^2 + 2s + 4)}$$

$$y_\infty = \lim_{s \to 0} s\,Y(s) = \frac{1}{4}\ \text{rad}$$

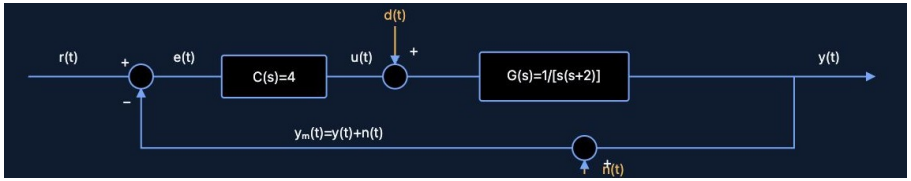
$$e_\infty = -0.25\ \text{rad}, \quad u_\infty = -1\ \text{rad/s}^2$$

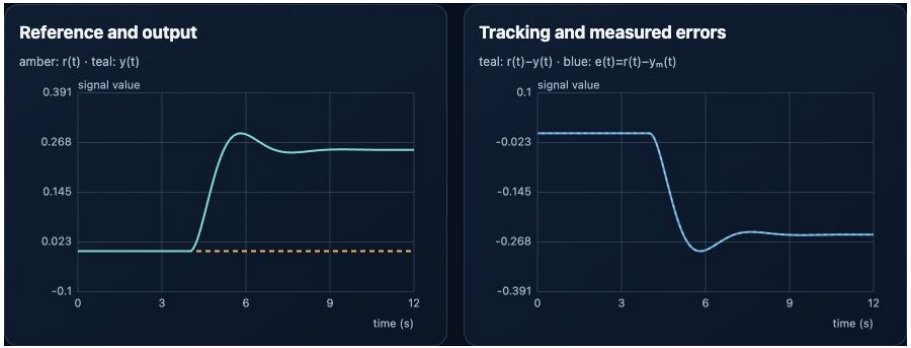
At rest, $0 = u_\infty + d_\infty$. A nonzero error is needed to produce the opposing P-control action.

~~This is load rejection, not reference-step tracking:~~ **~~Type 1 does not guarantee zero load offset~~**.

Live demo: load and measurement noise

OPEN LIVE DEMO ↗️





Feedback Explorer: Load disturbance

START Load disturbance: $r(t) = 0$, $K_c = 4$. A unit load begins at 4 s.

1 Predict the final $y(t)$, error and $u(t)$. Read all three plots. Double K_c: does the offset halve?

2 Select Measurement noise. Compare true error $r - y$ with measured error e. Watch control demand.

MATLAB alternative (Control System Toolbox)

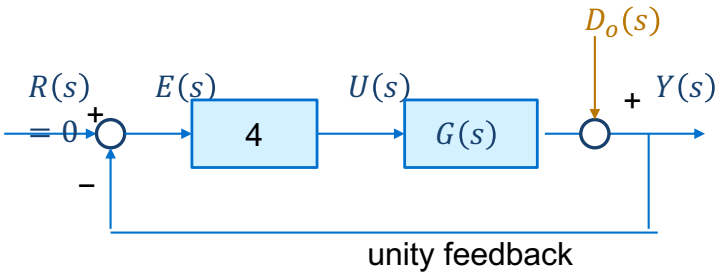
```
s=tf('s'); K=4; G=1/(s*(s+2));
T=feedback(K*G,1); H=feedback(G,K);
t=(0:.01:12)'; d=double(t>=4);
n=zeros(size(t)); % load-only case
y=lsim(H,d,t)-lsim(T,n,t);
e=-y-n; u=K*e;
figure; plot(t,[y,-y,e,u]); grid on
legend('y','r-y','e','u')
```

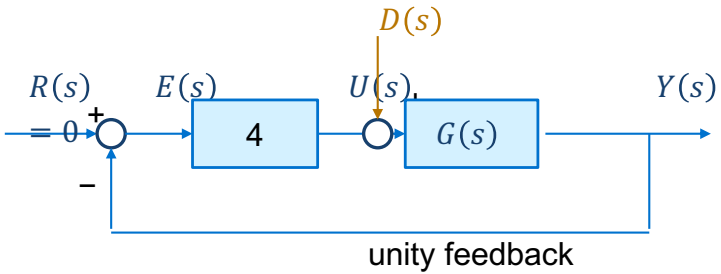
CHECK Unit input load: $y_\infty = 1/K = 0.25$ rad, $r - y \to -0.25$ rad, $u_\infty = -1$ rad/s². Noise enters the measured error through $e = r - y - n$.

OPTIONAL

Position loop C: change the disturbance entry

$G(s) = 1$ /[s(s+2)], $K = 4$; zero reference/noise and initial state. Compare final output for the two inputs below.





(a) Output displacement $d_o(t)$: 1 rad

$$Y = -4GY + D_o \;\Rightarrow\; (1+4G)Y = D_o$$

$$\frac{Y(s)}{D_o(s)} = S(s) = \frac{s(s+2)}{s^2+2s+4}$$

$$D_o(s) = \frac{1}{s}, \quad Y(s) = \frac{s+2}{s^2+2s+4}$$

$$y_\infty = \lim_{s\to 0} s\,Y(s) = 0$$

(b) Input load $d(t)$: 1 rad/s²

$$Y = G(-4Y + D) \;\Rightarrow\; (1+4G)Y = GD$$

$$\frac{Y(s)}{D(s)} = G(s)S(s) = \frac{1}{s^2+2s+4}$$

$$D(s) = \frac{1}{s}, \quad Y(s) = \frac{1}{s(s^2+2s+4)}$$

$$y_\infty = \lim_{s\to 0} s\,Y(s) = \frac{1}{4}\ \text{rad}$$

Input-load case b uses loop B (slide 10). Disturbance locations and units differ.

Gain K: error, damping and actuator demand

For the same motor, compare $K = 4$ and $K = 100$ before trusting the lower ramp error.

$$T(s) = \frac{K}{s^2 + 2s + K}, \quad \omega_n = \sqrt{K}, \quad \zeta = \frac{1}{\sqrt{K}}$$

Higher gain: smaller ramp error, lower damping

$$e_{\infty,\text{unit ramp}} = \frac{2}{K}, \quad M_p = 100e^{-\pi\zeta/\sqrt{1-\zeta^2}}\%$$

$K = 4$	$K = 100$
$\zeta = 0.50, \quad e_\infty = 0.50, \quad M_p = 16.3\%$	$\zeta = 0.10, \quad e_\infty = 0.02, \quad M_p = 72.9\%$

Actuator check: with a 1 rad initial error, $u(0^+) = K$ rad/s². If the actuator limit is 10 rad/s², $K = 100$ demands ten times the limit; the linear solution no longer applies.

These ideal loops remain stable.
Also check noise, delay and omitted resonances.

Question: does rejection depend on frequency?

Same motor, $K = 4$: compare a slow output disturbance with one at $\omega = 2$ rad/s.

Find $S(0)$, $|S(j2)|$ and the output amplitude for $d_o(t) = A\sin(2t)$ rad after $G(s)$.
Would an input load have the same amplitude ratio?

1 Calculate the sensitivity

$$S(0) = \lim_{s\to 0} \frac{s(s+2)}{s^2+2s+4} = 0$$

$$L(j2) = \frac{4}{j2(j2+2)} = -\frac{1}{2} - \frac{1}{2}j$$

$$S(j2) = \frac{1}{0.5 - 0.5j} = 1 + j$$

$$|S(j2)| = \sqrt{1^2+1^2} = \sqrt{2} \approx 1.414$$

2 Interpret the two input locations

Output-side displacement uses S:

$$A_y = |S(j2)|A = 1.414A \text{ rad}$$

Input load uses GS instead:

$$G(j2)S(j2) = \frac{1}{(j2)^2 + 2j2 + 4} = -\frac{j}{4}$$

$$A_y = 0.25A_d \quad (A_d \text{ in rad/s}^2)$$

DC limits describe zero-frequency behaviour only.
At 2 rad/s the output disturbance is amplified; the input load follows a different path.

Feedback benefits and costs

What large loop gain can improve

$$|L| \gg 1: \quad S = \frac{1}{1+L} \approx \frac{1}{L}$$

Small tracking error and reduced output-side disturbances in that band.

Lower relative plant-gain sensitivity, provided the closed loop remains stable.

For an input load, check GS—not S alone. The example still has a final offset of 1/K.

What it can cost

$$|L| \gg 1: \quad T = \frac{L}{1+L} \approx 1$$

Measurement noise reaches the output via −T.

Large commands can saturate the actuator.

In this motor, higher K reduces damping and increases overshoot.

Unmodelled resonances or delays can reduce stability margins.


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